Scientific Citation Retrieval

NDCG@10: 0.45 $\rightarrow$ 0.67

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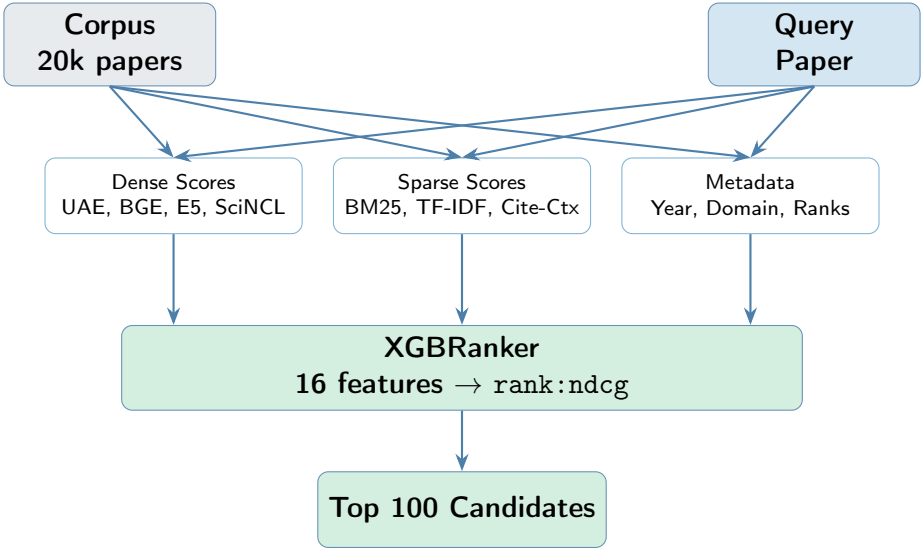
Pipeline Overview - Final Architecture

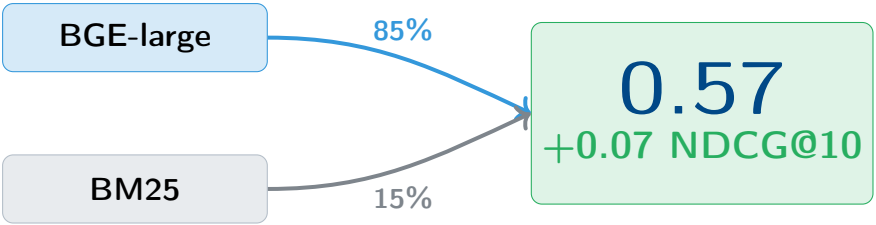


Ablation Study - Step 1: Baselines

TF-IDF	Dense MiniLM	BM25
(01) - provided	(02) - provided	(03) - ours
0.45	0.50	0.48
NDCG@10	NDCG@10	NDCG@10

Ceiling: sparse retrieval only $\rightarrow$ exact term matching

Ablation Study - Step 2: Score Fusion



Ablation Study - Step 3: Citation Context

Full text exploitation

"As shown by [1], neural networks can..."

"(Smith et al., 2020) demonstrated that..."

1. BM25 index on corpus **full text**
2. Extract citation sentences via `regex`
3. Score against full-text index
4. Fuse via 2D grid search

0.59

+0.02 NDCG@10

Signal is sparse, approaching plateau

Ablation Study - Step 4: Learning to Rank

Why LTR?

- Linear fusion can't capture interactions
- High UAE **and** high BM25 $\neq$ sum
- Need non-linear model

$\sim$16 features per (q, d):

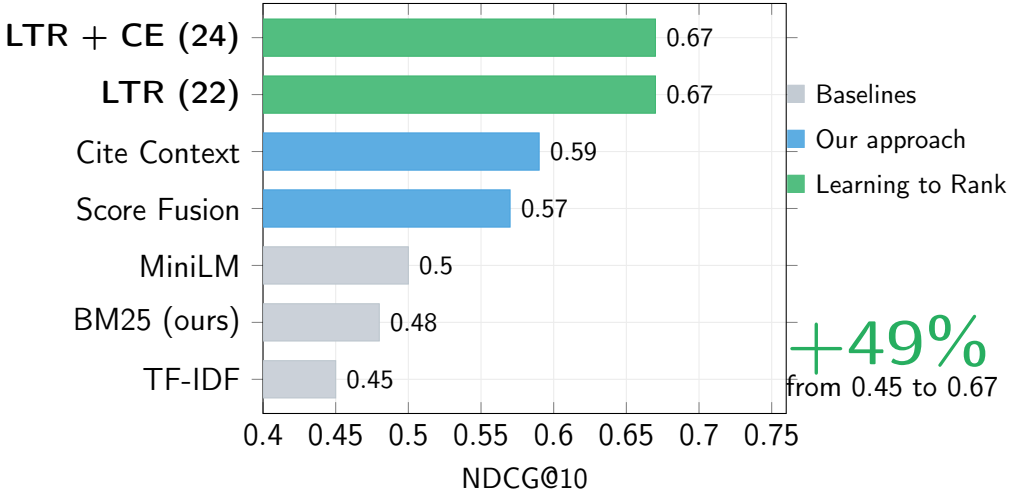
- Dense: UAE, BGE, E5, SciNCL
- Sparse: BM25 TA, fulltext, cite-ctx, TF-IDF
- Reciprocal ranks (6)
- Year proximity, domain match

XGBRanker

`objective: rank:ndcg`
`max_depth = 4`
`n_estimators = 200`

candidates: top-200 per system
eval: 5-fold GroupKFold

Results - Full Progression



Error Analysis - Failed Reranking Attempts

CE Reranking (33)

bge-reranker-v2-m3 on top-50

Blend: CE score + LTR rank

$<0.67 \downarrow$

CE already baked into LTR (script 24)
$\rightarrow$ no new information

LLM Reranking (34)

Qwen2.5-7B rates 0–10 per pair

top-20 only (cost)

$<0.67 \downarrow$

7B model too imprecise at this level
LTR already well-calibrated

Key lesson: Reranking on well-calibrated LTR only adds noise
The model already learned optimal signal weighting

Conclusion

$$\text{NDCG@10} \quad 0.45 \longrightarrow 0.67$$
$$(+49\%)$$

Key takeaways:

1. Better signals matter: MiniLM $\rightarrow$ BGE-large **+0.07**
2. Learning interactions $>$ linear fusion **+0.08**
3. Reranking on LTR $\rightarrow$ regression $\downarrow$